

Reproducibility Handbook

Environment, exact rerun, data governance, ethics, and protocol boundaries

Exact reproduction requires lawful access to all five datasets, including private Local Exp4, and the frozen Linux/CUDA environment. Bundle verification is not a substitute for scientific rerunning or governance approval.

Runtime	Linux x86-64; CPython 3.12.13; UV; PyTorch 2.6.0+cu124; TorchAudio 2.6.0+cu124
Formal scope	43 x 448 x 5 = 96,320 score-blind jobs
Preflight	172 / 172 CUDA construction and backpropagation checks
Data release	No raw EEG or trial predictions; pseudonymous derived metrics require review
License/authorship	Unresolved; no author or IRB identity inferred

Generated deterministically from the checksum-verified release bundle. No author identity was supplied or inferred.

Contents

Reproducibility guide 3

 Reproduction levels 3

 Frozen evidence identity 3

 1. Verify the distributed bundle 3

 2. Create isolated UV environments 3

 3. Run the software tests 4

 4. Acquire data and build private caches 4

 5. Choose new external run paths 5

 6. Execute the mandatory CUDA preflight 5

 7. Run or resume the exact grid 5

 8. Require the exact final audit 5

 9. Generate a new analysis publication 5

 Reproduction limitations 6

Data access, provenance, and redistribution boundary 6

 No EEG is distributed with this project 6

 Cohorts used by the verified common grid 6

 Authoritative public records and required citations 7

 Sealed cohorts not used in the common grid 7

 Local Exp4 release blocker 7

 Reproduction cache requirements 8

 Secure handling minimum 8

Ethics, privacy, and intended-use statement 9

 Research status 9

 Human-subject boundary 9

 Permitted present use 9

 Foreseeable harms and limitations 9

 Requirements before a patient-facing study 9

 Publication and release checklist 10

 Scientific claim boundary 10

Protocol boundaries and result-joining rules 11

 The non-negotiable rule 11

 Track matrix 11

 Common-grid comparison contract 12

 Coverage states 12

 Names are not join keys 12

 Participant and metric boundaries 12

 Claims allowed by the current common bundle 12

Source section: docs/REPRODUCIBILITY.md

Reproducibility guide

Reproduction levels

This project distinguishes three different claims:

- 1 **Bundle verification** checks source and result bytes without opening EEG.
- 2 **Software verification** recreates isolated UV environments and runs tests/model-construction checks without training the full grid.
- 3 **Scientific reproduction** lawfully reacquires the exact datasets, rebuilds provenance-bound caches, executes 96,320 GPU jobs, audits the complete score-blind tree, and regenerates the aggregate analysis.

Passing a lower level does not prove a higher one. A different dataset mirror, preprocessing choice, library lock, split, model roster, seed set, or GPU environment is a new experiment rather than an exact reproduction.

Frozen evidence identity

The distributed common result is under `results/common_grid_v6/`.

Evidence	Identity
Internal formal plan identity	d4dd852fc9e7b80c8c10c54df03d63ce0ae7e7555ee1b7abdee8d79dca28b61a
Raw plan.json SHA-256	a04c8f7ff9a8233c0dbdca986e0998a0e6c121e909eaafc69187d7c0cf93ef8
Preflight report.json SHA-256	39bef71293d33f343eb2d908f0c9bdc6cef06c9e457b98575ca06e86c80af4ef
Final final_audit.json SHA-256	010b44542157efb6ba5a95376198b9b691e9803ce3fe605e77cb5cdb9739d063
Analysis manifest.json SHA-256	960ca912d058a2ad3b60def4116dbba436f8ae9162916cde42a4ab265bf01f5d
Pre-result analysis-contract SHA-256	6eb975790bf7213e614159172d673b5e18ebda554f674e01c5b159099273393e
Analysis input-ledger SHA-256	3f68621a316a6a90cf18ce073e05f70b453a13fa7ef4407f66a90f54562dcbbf
Forensic-ledger SHA-256	962531ea433232d90d9869a2d78529f2e867cd7373b883dd78e8cf01ca31f626

The final audit has `exact_cartesian_complete=true`, 96,320 expected and complete records, zero missing, extra, failed, live-claim, stale-claim, or partial records, and a valid preflight. Six stale claim artifacts left by interrupted workers were preserved in quarantine and resolved by the forensic audit; they are not six failed scientific jobs.

The analysis directory is a sealed 17-file publication. `manifest.json` declares SHA-256 values for the other 16 files. It includes pseudonymous job-level, subject-seed-level, and subject-level derived metrics; it excludes raw EEG and trial-level predictions. Those participant-derived tables require governance review before public release. Generated convenience tables outside that directory are views; the sealed analyzer output remains the authority.

1. Verify the distributed bundle

From the project root:

```
python3 scripts/verify_release.py
```

The verifier checks required paths, the frozen core hashes, banned generated state, the sealed analysis manifest, and all entries in `results/SHA256SUMS`. It does not open EEG or recompute metrics from raw atomic predictions because those prediction packages are intentionally external to the clean release.

For an independent checksum pass on Linux:

```
cd results
sha256sum --check SHA256SUMS
```

2. Create isolated UV environments

Prerequisites:

- Linux x86-64;
- UV (the audited execution used UV 0.12.0);
- network access for the first lock-frozen synchronization, or a complete reviewed UV cache for offline synchronization;
- sufficient private disk space; and
- for formal execution, an NVIDIA driver compatible with the locked `torch==2.6.0+cu124` and `torchaudio==2.6.0+cu124` wheels.

The project pins CPython 3.12.13 in `.python-version`. The three environments are intentionally separate:

```
scripts/reproduce.sh setup runtime
scripts/reproduce.sh setup test
scripts/reproduce.sh setup docs
```

They create `.venv`, `.venv-test`, and `.venv-docs` from the same frozen `uv.lock`. No command uses `sudo`, system `pip`, or another user's environment. To select a particular UV executable:

```
UV_BIN=/absolute/path/to/uv scripts/reproduce.sh setup runtime
```

After `setup docs`, validate the sealed report inputs and generate the three deterministic PDFs. The default destination is `output/pdf/`; an alternate destination must be explicit:

```
scripts/reproduce.sh reports
scripts/reproduce.sh reports --output-dir /absolute/report-directory
scripts/reproduce.sh reports --validate-only
```

`--validate-only` creates no PDFs. A successful build does not replace the required page-by-page visual review before release.

After setup:

```
uv lock --check
UV_PROJECT_ENVIRONMENT="$PWD/.venv" uv pip check
UV_PROJECT_ENVIRONMENT="$PWD/.venv-test" uv pip check
```

Do not run `uv lock --upgrade`, `uv add`, an unfrozen sync, or an automatic environment repair after a formal plan is created. A changed dependency set requires a new project identity and run root.

3. Run the software tests

```
scripts/reproduce.sh test
```

The unit suite is not the 96,320-job benchmark. It exercises model geometry, loaders, deterministic training, strict schemas, source identity, the pinned TCFormer snapshot, GPU leases, crash/recovery behavior, exact auditing, and analysis publication. CUDA-specific cases require the reviewed Linux GPU environment. The frozen UV lock is Linux x86-64 only: macOS can run the standard-library bundle verifiers, but it cannot use the documented `setup/test` commands without creating a separately documented, nonformal environment.

4. Acquire data and build private caches

Read `docs/DATA_ACCESS.md` before downloading anything. The exact common grid requires all five datasets, including authorized Local Exp4 access. Create cache and raw-data roots outside this Git tree:

```
scripts/reproduce.sh prepare-data \
  /absolute/private/cache \
  /absolute/private/local-exp4
```

This requests only the opened development cohorts:

- BNCI2014-001 S1-S9;
- BNCI2014-004 S1-S9;
- Cho2017 S1-S52;
- PhysioNet MI S1-S54; and
- Local Exp4 participant codes 1, 3, 4, 5, 6, 7, 8, and 10.

It does not authorize access to the sealed cohorts. If the Local Exp4 path is omitted, the helper skips that private dataset; the resulting cache cannot run or be labelled as the exact five-dataset common grid.

Preserve acquisition URL/version, access date, original file hashes, exact terms, preprocessing identity, ordered channels, cache hashes, and split hashes. The formal planner validates those identities rather than accepting a shape-compatible replacement.

5. Choose new external run paths

Use previously unused absolute directories on a filesystem with at least 50 GiB free. Do not run inside `results/common_grid_v6`, a data cache, the source tree, another user's directory, or an earlier formal root.

```
export MI_CACHE_ROOT=/absolute/private/cache
export MI_RUN_ROOT=/absolute/new/common-grid-run
export MI_ANALYSIS_ROOT=/absolute/new/common-grid-analysis
```

The examples use task-specific variables intentionally; do not repurpose `HOME` or another system variable.

6. Execute the mandatory CUDA preflight

Select one approved idle physical GPU:

```
scripts/reproduce.sh preflight "$MI_CACHE_ROOT" "$MI_RUN_ROOT" 0
```

Preflight creates or validates the immutable plan and performs exactly 172 score-blind CUDA checks: 43 configurations against four canonical shape/class contracts. Each constructs the model, verifies finite logits and nonzero finite gradients, takes one AdamW step, and verifies a finite post-step forward pass. The attestation binds plan, source, environment, cache identities, GPU UUID, and shared GPU-lease receipt; it contains no test score.

Formal scripts set `PYTHONNOUSERSITE=1`, `PYTHONDONTWRITEBYTECODE=1`, `CUBLAS_WORKSPACE_CONFIG=:4096:8`, and the compact TCFormer source root.

7. Run or resume the exact grid

On a shared four-GPU workstation, the audited policy used GPUs 0-2 and left GPU 3 outside the launcher:

```
scripts/reproduce.sh run "$MI_CACHE_ROOT" "$MI_RUN_ROOT" 0,1,2
```

After an orderly pause, process interruption, or power loss, invoke the same command:

```
scripts/reproduce.sh resume "$MI_CACHE_ROOT" "$MI_RUN_ROOT" 0,1,2
```

The runner validates completed records and recovers only schema-recognized, identity-bound stale state. Do not edit claims, quarantine, records, plan, preflight files, or permissions by hand. Do not launch more than three project workers or include GPU 3 merely to shorten runtime on a shared host.

Read-only progress:

```
scripts/reproduce.sh status "$MI_RUN_ROOT"
```

The run is expensive and can take days depending on hardware, stalls, and shared-workstation load. Runtime measurements are workload-dependent even when predictions remain deterministic.

8. Require the exact final audit

```
scripts/reproduce.sh audit \
"$MI_RUN_ROOT" \
/absolute/new/common-grid-final-audit.json
```

Do not analyze unless the audit reports the exact Cartesian product complete, the preflight valid, and no missing, extra, failed, unsafe, partial, or live claim state. There is no available-case fallback.

9. Generate a new analysis publication

The destination must not exist:

```
scripts/reproduce.sh analyze \
"$MI_RUN_ROOT" \
"$MI_CACHE_ROOT" \
"$MI_ANALYSIS_ROOT"
```

Analysis reacquires the exclusive publication fence, revalidates the complete prediction tree, joins labels, computes participant-level metrics and 100,000 deterministic bootstrap resamples, and publishes a new sealed 17-file directory. Compare its plan, analysis-contract, input-ledger, environment, source, and manifest identities before comparing numbers with the distributed analysis.

Reproduction limitations

- Raw atomic prediction records are not distributed, so the sealed aggregate bundle can be checksum-verified but not independently rescored without rerunning or obtaining the authorized external result root.
- Local Exp4 is private and its release authorization is not documented in this project. A public-data-only rerun is scientifically useful but is not the same fixed five-dataset suite.
- Exact dataset-file licenses and hashes must follow the bytes actually acquired; MOABB metadata is not a substitute.
- A driver, GPU, library, compiler, or filesystem change can alter runtime identity and possibly numerical bytes. Report the new identity rather than relabelling it as the original run.
- The sealed confirmation cohorts remain unopened. Reproduction of development results does not authorize confirmation access.

Source section: docs/DATA_ACCESS.md

Data access, provenance, and redistribution boundary

No EEG is distributed with this project

This repository contains code, configuration, documentation, and a sealed development analysis. That analysis includes pseudonymous job-level, subject-seed-level, and subject-level derived metric tables. It does not include raw EEG, transformed/harmonized caches, trial-level predictions, direct participant identifiers, demographic/clinical metadata, features, embeddings, checkpoints, or private acquisition manifests.

Pseudonymous participant keys and metrics are still participant-derived data. Their inclusion in an internal release candidate is not authorization for public disclosure. In particular, the Local Exp4 rows in `job_metrics.csv`, `subject_seed_metrics.csv`, and `subject_metrics.csv` require a written governance decision before this bundle may be published.

Reproduction therefore requires the researcher to obtain each public dataset from its authoritative record, comply with its terms, create a private cache, and retain a provenance manifest. Local Exp4 is not public and may be used only under its controlling institutional authorization.

Filtering, resampling, rereferencing, epoching, pseudonymizing, or converting files does not erase the original dataset's license, confidentiality, or human-subject obligations.

Cohorts used by the verified common grid

All 132 participant caches in the completed common grid were already opened during model development. The resulting scores are development evidence, not independent confirmation.

Dataset	Common-grid cohort and task	Frozen evaluation boundary	Access/redistribution status
Local Exp4	8 pseudonymous participants; binary left/right imagery	Four chronological recordings: first two train, third validation, fourth prediction-only test	Private. No supplied document establishes the governing IRB/consent/data-use decision. Do not distribute any participant-derived artifact without written approval.
BNCI2014-001 / BCI Competition IV 2a	9 participants; four classes: left hand, right hand, feet, tongue	Official evaluation session held out; final source-training run within the training session is validation	Official BNCI record reports CC BY-ND 4.0. Do not distribute transformed caches without written permission or a documented legal determination.
BNCI2014-004 / BCI Competition IV 2b	9 participants; binary left/right; three supplied bipolar derivations	Official chronological/session holdout	Official BNCI record reports CC BY-ND 4.0. The three signals are bipolar derivations, not interchangeable with C3/Cz/C4 point electrodes. Do not distribute transformed caches.
Cho2017	52 participants; binary left/right	Five rotating, class-stratified acquisition-order block folds	Publicly accessible through the dataset DOI, but the exact license attached to the acquired bytes must be captured. Current records are not sufficiently uniform to authorize redistribution from this project.
PhysioNet EEG Motor Movement/Imagery v1.0.0	Participants S1-S54; imagined unilateral-fist runs 4, 8, and 12; binary left/right	Three rotating complete-run folds	Official files are labelled ODC-By 1.0. This project nevertheless does not bundle originals or transformed caches. S55-S109 remain sealed confirmation subjects.

The common public preprocessing profile is 4-40 Hz, 0.5-3.0 s as a half-open interval, 128 Hz, 320 samples, epoch demeaning, and symmetric common-average reference except for the supplied BNCI2014-004 bipolar signals. The normal profile uses a declared 21-channel montage; BNCI2014-004 retains three bipolar derivations. Local Exp4 uses its separate, task-valid 0.0-2.0 s half-open window, 128 Hz/256 samples after resampling from 125 Hz, and common-average reference over 15 point-EEG channels.

Source-only channel scaling is fitted inside each split. The selection scaler uses training rows only; the final-refit scaler uses train plus validation; test rows never determine scaling.

Authoritative public records and required citations

- **BNCI2014-001:** obtain from the official [BNCI data-set registry](https://bnci-horizon-2020.eu/database/data-sets) [https://bnci-horizon-2020.eu/database/data-sets]. Cite the registry and Tangermann et al., "Review of the BCI Competition IV," *Frontiers in Neuroscience* (2012), DOI [10.3389/fnins.2012.00055](https://doi.org/10.3389/fnins.2012.00055) [https://doi.org/10.3389/fnins.2012.00055].
- **BNCI2014-004:** obtain from the same official BNCI registry. Cite the registry and Leeb et al., "A brain-computer interface based on imagined hand movements," *IEEE TNSRE* (2007), DOI [10.1109/TNSRE.2007.906956](https://doi.org/10.1109/TNSRE.2007.906956) [https://doi.org/10.1109/TNSRE.2007.906956].
- **Cho2017:** use the supporting-data record [GigaDB 100295](https://doi.org/10.5524/100295) [https://doi.org/10.5524/100295] and cite Cho et al., "EEG datasets for motor imagery brain-computer interface," *GigaScience* (2017), DOI [10.1093/gigascience/gix034](https://doi.org/10.1093/gigascience/gix034) [https://doi.org/10.1093/gigascience/gix034]. Record the exact mirror, version, terms, and hashes actually used.
- **PhysioNet:** use [EEG Motor Movement/Imagery Database v1.0.0](https://physionet.org/content/eegmmidb/1.0.0/) [https://physionet.org/content/eegmmidb/1.0.0/], DOI [10.13026/C28G6P](https://doi.org/10.13026/C28G6P) [https://doi.org/10.13026/C28G6P], and preserve the official page's requested citations and ODC-By notice.

MOABB is an acquisition interface, not the licensor. Its metadata does not replace the exact terms served with the acquired files.

Sealed cohorts not used in the common grid

The following are not missing common-grid results. They were deliberately reserved from the opened development suite:

- PhysioNet S55-S109;
- all 54 Lee2019/OpenBMI MI participants;
- all 10 Weibo2014 participants; and
- all 4 Zhou2016 participants.

Do not download, inspect, cache, tune on, or score these cohorts merely to complete a table. Opening them requires a prespecified confirmation protocol, a frozen model/configuration, rights review, and authorized unsealing. Once opened, they cannot continue to be described as untouched confirmation data.

Local Exp4 release blocker

The repository does not establish the institution, protocol title or number, approval/exemption status, consent language, participant population, recruitment procedure, data controller, or permitted artifact classes for Local Exp4. No such fact may be inferred from the project's assistive-BCI motivation.

Before any external sharing or manuscript release, the principal investigator, data steward, and institutional privacy/IRB authority must document whether the approved boundary permits:

- this secondary machine-learning analysis;
- publication of aggregate statistics;
- external collaboration and public code;
- per-participant or trial-level outputs;
- trained weights or embeddings;
- retention, withdrawal, correction, and deletion procedures; and
- commercial or assistive-device use.

Until then, keep every Local Exp4-derived artifact, including the bundled pseudonymous metric rows and any weights, within the approved environment. Even dataset-level aggregate values require the research team's publication authorization. If participant-level release is not permitted, generate a new privacy-minimized publication and new manifest/checksums from an authorized pipeline; do not delete or edit rows inside the sealed analysis directory.

Reproduction cache requirements

Use an external private cache root; never place it inside the Git project. For every subject, preserve a machine-readable manifest containing:

- 1 canonical dataset identifier and version;
- 2 exact acquisition record/URL and access date;
- 3 acquisition library and locked version;
- 4 original filename, byte size, and SHA-256;
- 5 exact license or terms captured with those bytes;
- 6 preprocessing schema, ordered channels, events, coordinate convention, and split identity;
- 7 cache array and metadata digests emitted by the loader; and
- 8 an aggregate manifest digest referenced by the run plan.

Missing identity information is a provenance failure. Do not substitute a nearby mirror or silently rebuild a cache after a plan is frozen.

Secure handling minimum

- Store raw and derived EEG only on institutionally approved encrypted storage with least-privilege access.
- Keep participant-code linkage outside the analysis project and under a different access boundary.
- Never put credentials, tokens, private hostnames, or identifying paths in manifests intended for release.
- Treat EEG, predictions, embeddings, and checkpoints as potentially identifying; review re-identification and membership-inference risk.
- Record access, exports, incidents, withdrawals, retention, and destruction according to the controlling protocol.
- Release dataset-level or participant-level tables only after checking small-cell, outlier, linkage, and singling-out risks and obtaining the required approval.

The longer source audit in `docs/DATASETS_AND_LICENSES.md` records additional dataset-level caveats. If it conflicts with newly captured authoritative terms, stop release work, preserve both records, and obtain a documented decision rather than choosing the more permissive interpretation.

Source section: docs/ETHICS_AND_PRIVACY.md

Ethics, privacy, and intended-use statement

Research status

This is research software for offline motor-imagery EEG classification. It is not a medical device, clinical decision system, communication aid approved for patient use, or evidence that an assistive intervention benefits disabled or paralyzed people.

The motivating goal is important, but the verified common-grid evidence comes from previously opened development cohorts and primarily healthy volunteers. No included experiment establishes safety, effectiveness, usability, calibration, or quality-of-life benefit for the intended clinical population.

Human-subject boundary

EEG is human-subject data and can remain sensitive after names are removed. Signals, session patterns, rare phenotypes, linked metadata, model embeddings, and trained weights may permit singling out, membership inference, or re-identification. Pseudonyms are not anonymization.

For the private Local Exp4 cohort, the supplied project materials do not state an IRB protocol number, approval/exemption status, consent version, recruitment population, demographics, or data-use authorization. This project therefore makes no claim about those facts. External release is blocked until the responsible institution supplies and approves them. See [docs/DATA_ACCESS.md](#).

Public availability of the four external datasets does not remove the duties to follow their terms, cite them, minimize data, protect participants, and respect any withdrawal or use restrictions.

Permitted present use

Subject to the user's institutional authorization and each dataset's terms, the present defensible use is:

- offline methods research;
- reproducibility checks against the frozen development protocol;
- aggregate, protocol-labelled comparison of the 43 common-recipe models; and
- engineering evaluation of runtime, memory, and calibration.

The software must not be used to control mobility, communication, stimulation, clinical care, or another safety-critical system without a separate approved study and an appropriate real-time safety architecture.

Foreseeable harms and limitations

- **False activation and non-response:** offline balanced accuracy does not measure asynchronous false commands, abstention, dwell time, fatigue, or failure during real-world assistive use.
- **Population shift:** performance can change with disability, paralysis, medication, lesions, implants, age, skin/hair characteristics, language, attention, fatigue, montage, amplifier, or care setting.
- **Unequal performance:** aggregate means can conceal participants for whom the decoder performs at or below chance. No demographic fairness analysis is available because approved demographic metadata were not supplied.
- **Calibration:** the descriptive common-grid leader had worse aggregate calibration than TCFormer in this development analysis; high confidence must not be interpreted as clinical reliability.
- **Privacy leakage:** participant-level outputs and weights can reveal more than aggregate tables.
- **Automation bias:** a high development score may encourage unsafe trust, especially when the observed winner was selected from 43 configurations.

Requirements before a patient-facing study

At minimum, a responsible clinical and engineering team must establish:

- 1 IRB/ethics approval and consent specifically covering the target population, intended task, device interaction, data flows, and failure modes;
- 2 participatory design with disabled users, caregivers, clinicians, and accessibility specialists, including compensation and burden review;
- 3 a prospective protocol with a frozen model and independent participants;
- 4 real-time metrics: false activations per unit time, latency, information transfer, abstention, recalibration burden, fatigue, session drift, and participant-specific failure rates;
- 5 human override, fail-safe states, command confirmation, monitoring, and an incident-response plan;
- 6 security threat modeling, access controls, encryption, audit logging, and a retention/deletion schedule;
- 7 subgroup and worst-case analysis with a plan for non-users and declining performance;
- 8 clinical, regulatory, and institutional review of the claimed intended use; and
- 9 transparent communication that participation or poor decoder performance is not a measure of cognition, effort, competence, or eligibility for care.

Publication and release checklist

Before releasing any artifact, obtain a written decision for each row:

Artifact class	Current status	Required authority
In-house source code	License unresolved	Copyright holders and institutional release authority
Aggregate public-dataset statistics	Development-only; dataset citations required	Research team and dataset terms
Aggregate Local Exp4 statistics	Included in the internal sealed analysis; public-release authorization not documented here	PI, data steward, and IRB/privacy authority
Raw EEG or transformed caches	Not included and not authorized by this project	Dataset terms plus institutional/data-controller approval
Pseudonymous job/subject-seed/subject metrics	Included in the sealed analysis; still participant-derived and not approved here for public release	Explicit privacy/IRB/data-use and disclosure-control approval
Trial-level predictions	Not included	Explicit privacy/IRB/data-use approval
Embeddings or trained weights	Not included; potentially participant-derived	Privacy, IP, dataset, sponsor, and IRB/data-use review
Participant demographics	Not supplied	Consent, minimization, and disclosure-control review

Do not backfill missing approvals, demographics, disability status, funding, or consent language from assumptions. A manuscript should report unknowns as unknowns until the responsible authority supplies them.

Scientific claim boundary

The completed common grid supports one narrow statement: under one frozen training recipe and five opened development datasets, the observed equal-dataset balanced-accuracy leader was an in-house CardinalFBC configuration. The descriptive uncertainty interval for its difference from TCFormer included zero. The result is not independent confirmation, a global state-of-the-art result, proof of novelty, or evidence of clinical benefit.

Source section: docs/PROTOCOL_BOUNDARIES.md

Protocol boundaries and result-joining rules

The non-negotiable rule

Numbers may share a report, but they may be ranked or contrasted as peers only when model, inputs, participants, folds, seeds, training recipe, selection, reset/refit, and test-use contracts match. A result from one row below cannot fill a missing cell in another.

The only completed five-dataset architecture leaderboard distributed with this release is the **43-model common raw-trial recipe**. "All models" in that context means all 43 members of the frozen common roster, not every registry entry, historical procedure, author-recipe reference, transfer condition, control, or experimental candidate in the repository.

Track matrix

Track	Scientific question	Eligibility and unit	Evidence status in this release	Comparison boundary
Common raw-trial recipe	How do 43 fixed architecture/configuration factories compare under one recipe?	Five opened datasets; 448 participant/fold units; five seeds; 96,320 jobs	Complete, exactly audited, and bundled	Rank only the 43 common rows. TCFormer is the prespecified context comparator.
Gauge Gate 1	Does one frozen experimental continuation justify disjoint Gate 2?	17 prespecified opened participants, one candidate, bound common-reference predictions	Complete negative gate; not in common roster	Compare only under its gate contract. Failed breadth criterion forbids promotion.
CHSD screens/robustness amendment	Do complex HSD representations clear frozen promotion criteria?	Separate screened and disjoint-subject subsets	Negative development evidence documented; not registered/common	Report as a screen, never as a common-grid row.
Author-recipe references	How do adapted TCFormer and FBCNet behave under training recipes closer to their publications?	Same eligible datasets but method-specific optimization; 4,480 planned records	Formal runner is included; no completed author-recipe result is distributed or claimed by this release	Compare the two fixed author-recipe references only inside their own table. Do not substitute these values for common TCFormer/FBCNet.
GeoAdapt harmonized-v2	How do geometric architectures/procedures behave with their SPD inputs?	Binary-eligible L/B/C/P; 6,495 planned records	Runner/protocol work is included, but the harmonized-v2 grid is unrun and no score is distributed	Compare only matching GeoAdapt conditions and eligible datasets.
Local binary procedures	How do CAMEO, HemiParity, PARITY-Fuse, and ORBIT-v3 behave with paired-view/procedure-specific training?	Binary L/B/C/P; BNCI2014-001 is principled N/A; 8,780 planned records	Frozen Local Exp4 procedure artifacts exist historically; the all-dataset formal procedure grid is unrun	Keep one separate procedure table; historical Cho/legacy-A scores use different protocols.
Deterministic classical controls	How do tangent/Riemannian/EA-FBCSP controls perform without stochastic seed aliases?	Five datasets; one seedless record per participant/fold; 1,344 planned records	A separate CPU runner is included, but no completed control grid is distributed or claimed	No invented five-seed replication; compare within the control table or exact paired contexts.
HemiQ harmonized-v2	How does the dataset-specific reflection-odd field behave on supplied BNCI2014-004 bipolar signals?	BNCI2014-004 only; 45 planned records	Historical development/one-time confirmation evidence exists; the harmonized-v2 grid is unrun and raw historical predictions are excluded from this clean bundle	N/A elsewhere. Never treat the three derivations as point electrodes.
Native CardinalFBMS transfer	Does a source checkpoint help on native Cho/PhysioNet target montages?	Cho S16-S52 and PhysioNet S1-S54; five conditions; 8,675 records	A prior complete artifact was independently verified; not bundled into the common ranking	Compare only the five matched transfer conditions.
Native CardinalFBC transfer	Same transfer question for predecessor FBC conditions	Same native target cohorts; four conditions; 6,940 total when complete	Only limited predecessor coverage is documented; full matched table is incomplete	No available-case aggregate and no substitution from common CardinalFBC.
CardinalSplineDualView	Does fixed geometry transport plus native/canonical fusion help target adaptation?	Native Cho/PhysioNet transfer cohorts	Prototype only; no promoted aggregate writer/result	Blocked, not zero and not a common cell.
Sealed confirmation	Does a frozen selected method replicate on untouched cohorts?	PhysioNet S55-S109, Lee2019, WeiBo2014, Zhou2016 under a future protocol	Not run and not authorized by this release	Must remain absent until lawful, prespecified unsealing.

"Not bundled" is not a performance statement. It means that this clean common-grid result package does not supply a canonical same-protocol aggregate for that track. Consult the named track's own immutable artifact and audit before making any claim about

completion.

Common-grid comparison contract

The common roster is immutable. It contains 28 in-house configurations and 15 external configurations. Adding Gauge, CHSD, a procedure model, a new variant, or an author-recipe result would change the cardinality and invalidate the formal plan identity.

The observed equal-dataset leader is selected descriptively from this roster. The winner-versus-TCFormer interval is not confirmatory because the winner was selected using the same outcomes. No p-value or null-hypothesis decision is reported. A small numerical lead or an interval that crosses zero must not be restated as equivalence, non-inferiority, or superiority.

Coverage states

Use exactly these meanings when building broader tables:

- **completed**: the exact named track has a validated result for the cell;
- **planned_incomplete**: the cell is eligible and prespecified but the exact track is not complete;
- **blocked**: the scientific cell may be meaningful, but an adapter, configuration identity, checkpoint, provenance decision, or approval is missing; and
- **not_applicable**: the method cannot answer that cell's frozen task, such as a binary-only parity procedure on four-class BNCI2014-001.

Never encode planned, blocked, or not-applicable cells as zero. Never impute them into an overall average. Do not mark a cell completed using a result from a differently preprocessed, differently split, historical, author-recipe, deployment, transfer, or candidate-screen artifact.

Names are not join keys

Some display names recur across tracks. Common TCFormer and author-recipe TCFormer are intentional aliases for one architecture under different training procedures. Common FBCNet and author-recipe FBCNet have the same relationship. CardinalFBMS from scratch in the common recipe is not the pretrained CardinalFBMS transfer procedure. ORBIT-v1-v5 share one classifier class but depend on distinct configurations.

Join results only with a stable registry ID plus the exact track and artifact digest. Do not join by display name, Python class, family name, runtime condition, or nearest available score.

Participant and metric boundaries

- The participant is the independent unit. Folds, trials, and seeds are repeated measures.
- Common overall balanced accuracy is an equal average of five dataset means, not a trial-pooled micro score.
- Trial-micro accuracy, NLL, Brier, and ECE may be shown separately, but they weight cohorts by trial count and repeated seeds.
- Four-class and binary class semantics differ. Cross-dataset macro AUROC and other pooled classification metrics are intentionally undefined in the trial-micro table.
- Runtime reflects shared-workstation conditions and is an engineering measurement, not a scientific performance tie-breaker.

Claims allowed by the current common bundle

It is accurate to say that the exact common grid completed 96,320 jobs with no missing, failed, extra, partial, or live-claim jobs, and that `cardinal_fbc_compactdyn_scale025_extended` was the observed descriptive leader under the frozen common recipe.

It is not accurate to call that model independently confirmed, globally best, state of the art, clinically validated, proven useful for disabled or paralyzed people, or definitively superior to TCFormer. It is also not accurate to call Gauge a successful promoted model: Gauge failed its frozen Gate 1 even though its overall subset mean was numerically favorable.